Annotation Guide to Write Phenoscript Descriptions

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Introduction

Phenoscript uses **Entity-Quality (EQ) syntax** as the foundation for all phenotypic statements, allowing precise description of morphological traits through ontologies.

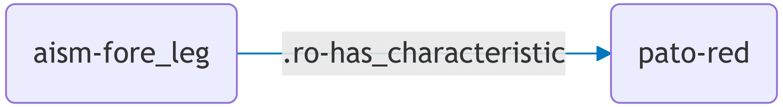
This approach builds upon previous standardized frameworks, such as those documented in the Phenoscape Character Annotation Guide.

Semantic statements

In the **Entity–Quality (EQ)** model, a morphological trait is expressed as a triplet linking a **structure** to a **quality**.

It is convenient to think of them as **knowledge graphs** composed of two nodes connected by an edge.

```
# Fore leg is red
aism-fore_leg .ro-has_characteristic pato-red;
```



We extend this syntax to also represent relationships between various combinations of node types (*e.g.*, two entities).

Nodes (`N`)

Four node types:

Type	Description	Example
Class node	Maps to an ontology class → creates an OWL individual	`uberon-male_organism`
Integer node	Whole number (counts)	`2`
Real node	Decimal number (measurements)	`2.0`
String node	Free text in quotes	`'Helictopleurus sicardi'`

```
uberon-male_organism .bfio-has_part aism-protibia;
uberon-male_organism .iao-has_measurement_value 2.0;
uberon-male_organism .rdfs-label 'Helictopleurus sicardi';
```

Nodes: classes

- **Entity**: something that can be referred to as a distinct thing
  - **Material**: protibia, anterior region, etc.
  - **Immaterial**: anatomical line, body axis, etc.
- **Quality**: an intrinsic feature of an entity
  - **Value**: red, curved, flattened, etc.
  - **Relational**: distal to, fused with, etc.

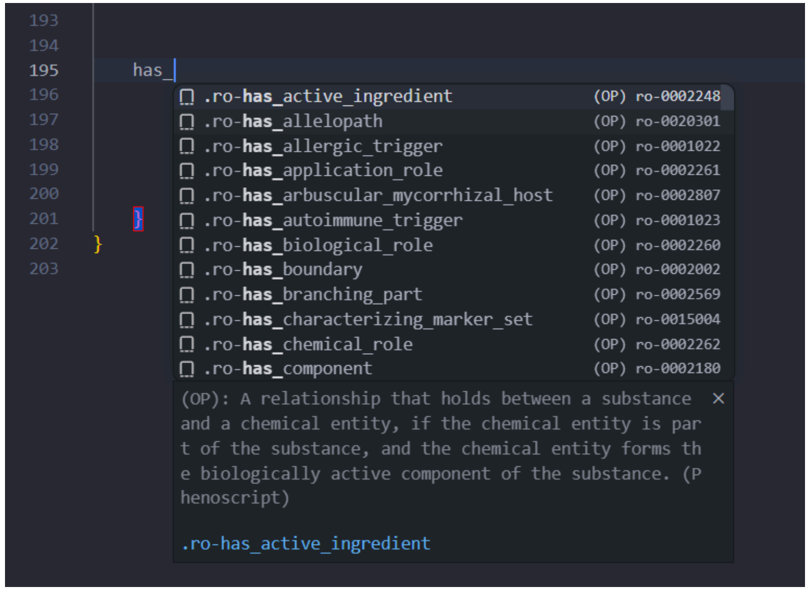
Edges ('.E')

They represent **relationships** between nodes (*e.g.*, part of, has characteristic, encircles, etc.).

Three edge types corresponding to OWL property types:

Type	Abbreviation	Connects	Participates in reasoning?
Object property	**OP**	class → class	✅
Data property	**DP**	class → string or number	✅
Annotation property	**AP**	any → any	❌

VS Code snippets label each edge as **(OP)**, **(DP)**, or **(AP)**.



> Phenoscript does **not** enforce correct edge usage — care is required.

Semantic statements — Rules

Rules:

- Terms are written with their prefixes (`aism-`, `pato-`, `ro-`), which identify the source ontology — inserted automatically when selecting a snippet
- Edges start with a dot `.` — inserted automatically when selecting a snippet
- Each statement must **begin and end with a node**

```
# Correct — ends with a node
N .E N .E N;

# Incorrect — ends with an edge
N .E N .E;
```

- Statements can be as long as needed
- Every statement must end with a **semicolon** `;`

Comments are marked with `#` and are ignored by the converter.

Building phenotypic statements

Simple composition:

```
this .bfo-has_part aism-protibia;
```

Complete Entity-Quality statement:

```
this .bfo-has_part aism-protibia .ro-has_characteristic pato-red;
```

How to know what descriptive terms can be used?

All available terms are found in the ontologies (UBERON, PATO, BFO, RO, AISM, etc.).

Snippets:

Phenoscript snippets allow to look for terms and insert them in the description very easily:

Ontology term lookup tables:

Terms	Table
Qualities (PATO)	pato-qualities/table.html
Insect anatomy (AISM + COLAO)	aism-terms/table.html

Personalized tags: `:`

When the same individual appears on multiple lines, use a **tag** to identify all occurrences as the same individual.

```
node:tag_name
```

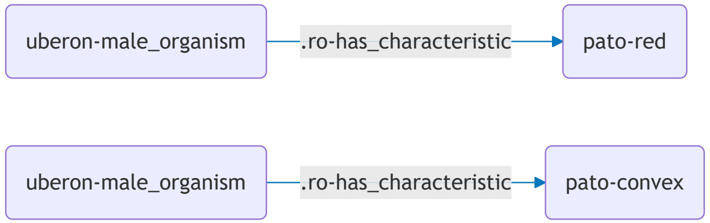
Without a tag, each occurrence is treated as a **different individual** — causing logical errors.

Use the **id tag** snippet to generate a unique random tag automatically.

Tags: Without vs With

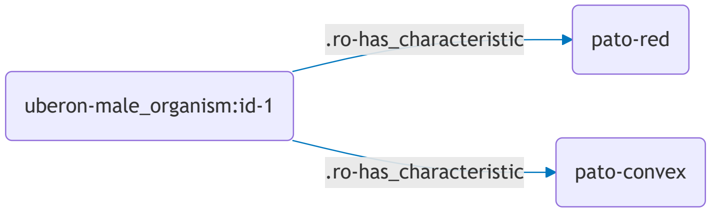
Without tag — two individuals (incorrect)

```
uberon-male_organism >> pato-red;
uberon-male_organism >> pato-convex;
```



With tag — one individual (correct)

```
uberon-male_organism:id-1 >> pato-red;
uberon-male_organism:id-1 >> pato-convex;
```



Node lists: `(N1, N2, N3)`

A **node list** links one node to multiple targets in a single statement:

```
N1 .E (N2, N3, N4)
```

```
# Assign two qualities at once:
uberon-male_organism >> (pato-red, pato-convex);
```

Equivalent to writing each statement separately:

```
uberon-male_organism:id-1 >> pato-red;
uberon-male_organism:id-1 >> pato-convex;
```

Node lists help keep descriptions concise when one structure has several qualities.

Shortcut: `this`

`this` is used to reference to the specimen:

```
# Organism is red
this >> pato-red;
# same as: uberon-male_organism:carabus_nemoralis >> pato-red;
```

Structure of PhenoScript descriptions

OTU structure

All statements must be placed inside an **OTU** (Operational Taxonomic Unit) block.

```
OTU = {
  DATA = {}    # specimen metadata: taxonomy, catalogue number, labels
  TRAITS = {}  # morphological trait statements
}
```

- `DATA` — who and what the specimen is
- `TRAITS` — what traits the specimen has

Best practice: one species per `.yphs` file, one OTU per specimen.

Use the snippet **`tmp: Insert OTU`** to insert the block template.

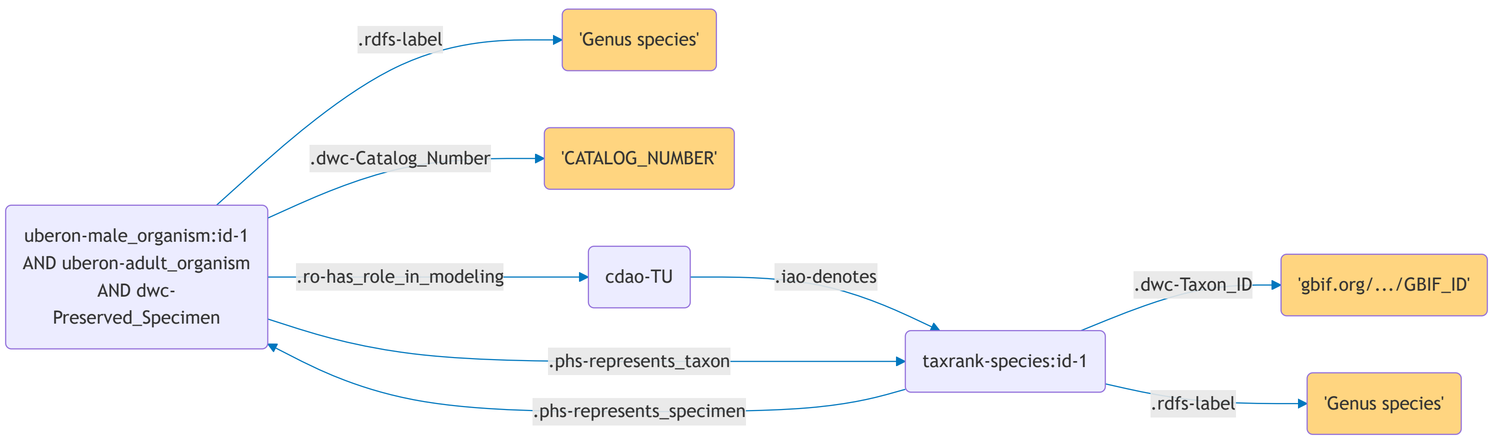
YAML blocks: motivation

Expressing specimen metadata as raw Phenoscript is verbose:

```
DATA = {
  uberon-male_organism:genus_species[this = True, linksTraits = True,
      cls = 'uberon-adult_organism', cls = 'dwc-Preserved_Specimen']
      .rdfs-label '_ORG_Genus species';
  uberon-male_organism:genus_species .dwc-Catalog_Number 'CATALOG_NUMBER';
  uberon-male_organism:genus_species .ro-has_role_in_modeling cdao-TU
      .iao-denotes taxrank-species:yml-2db3c3;
  taxrank-species:yml-2db3c3 .dwc-Taxon_ID_taxonID
      'https://www.gbif.org/species/GBIF_ID';
  taxrank-species:yml-2db3c3 .rdfs-label 'tax_Genus species';
  taxrank-species:yml-2db3c3 .phs-represents_specimen
      uberon-male_organism:genus_species;
  uberon-male_organism:genus_species .phs-represents_taxon
      taxrank-species:yml-2db3c3;
}
```

YAML blocks: motivation (graph)

The verbose DATA block above produces this knowledge graph:



YAML block for compactness

The same data in a YAML block (*Carabus nemoralis*, voucher `Luomus:123`):

```
OTU = {
  DATA = {
    #>>>YAML described_species
    described_species:
        .id: carabus_nemoralis
        .rdfs-label: 'Carabus nemoralis'
        .gbif_id: 'https://www.gbif.org/species/8056040'
        .dwc-Catalog_Number:
            - 'Luomus:123'
        .is_a:
            - uberon-male_organism
            - uberon-adult_organism
            - dwc-Preserved_Specimen
    #<<<YAML
  }
  TRAITS = {
    this >> pato-red;
    this >> pato-convex;
  }
}
```

YAML block for a new species

The same data in a YAML block (*Carabus nemoralis*, voucher `Luomus:123`):

```
#>>>YAML new_species
new_species:
    .id: genus_species
    .rdfs-label: 'Genus species'
    .zoobank_id: 'http://zoobank.org/ZOOBANK_ID'
    .dwc-Parent_Name_Usage_ID: 'https://www.gbif.org/species/GBIF_ID_of-Parent-Genus'
    .dwc-Catalog_Number:
        - 'CATALOG_NUMBER'
    .is_a:
        - uberon-male_organism
        - uberon-adult_organism
        - dwc-Preserved_Specimen
#<<<YA
```

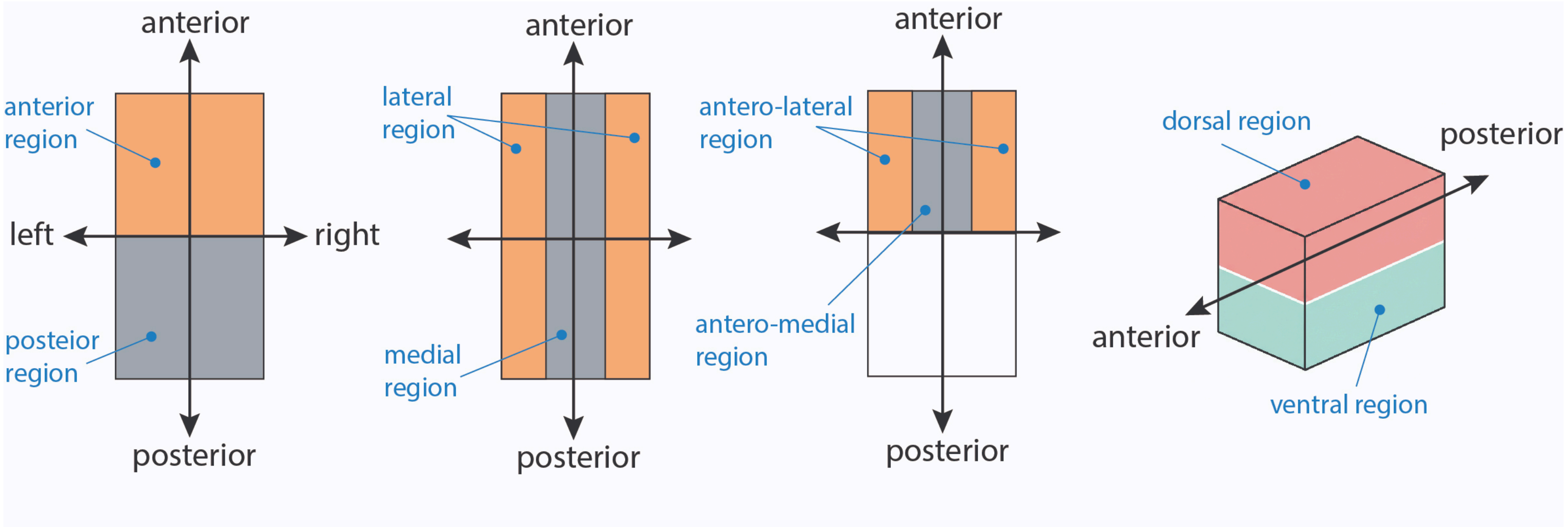
YAML blocks: key points

- `.yphs` files support YAML blocks; `.phs` files do not
- Use **YPHS → PHS** (right-click) to see how the block expands
- Always add a **GBIF ID** to link the taxon to a global registry
- Snippet `tmp: Described species` — for known species
- Snippet `tmp: New species` — for species new to science (requires ZooBank ID + parent genus GBIF ID)

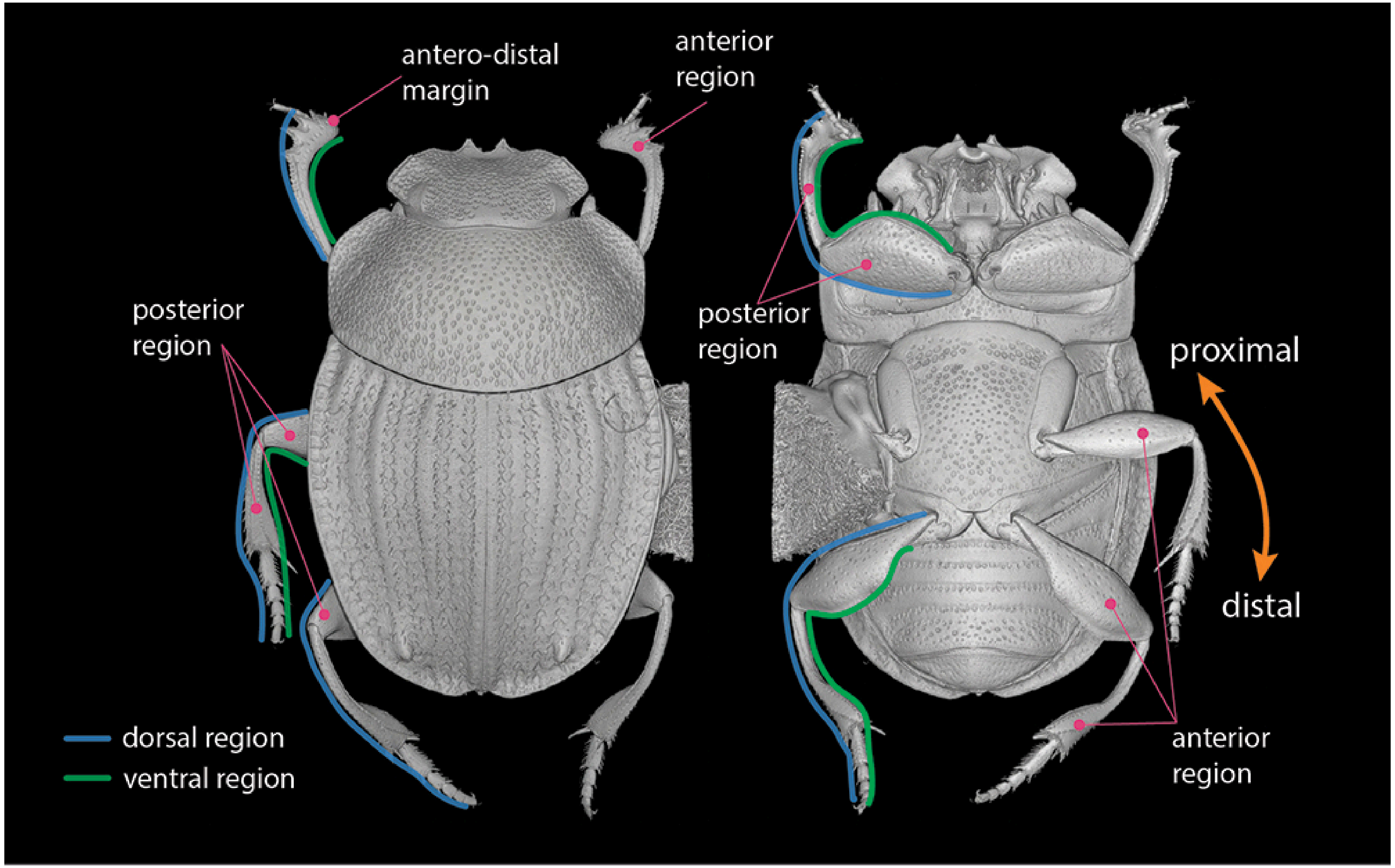
Positional terminology

Positional terms

Positional terms must refer to the organism's **main body axes**, not relative axes.



Example: In some Coleoptera, protibial teeth may appear "lateral" traditionally but are actually **dorsal** on the protibia.



Main body parts:

- anterior, posterior region/margin
- dorsal, ventral
- lateral, medial

Appendages:

- anterior, posterior region/margin
- dorsal, ventral
- proximal, distal

... and their combinations: antero-lateral, antero-medial, ventro-proximal, etc.

⚠️ **Do NOT use** basal or apical.

Body regions and margins

- Use **region** (`bspo-anatomical_region`)
- Use **margin** (`bspo-anatomical_margin`)

⚠️ Do not use `anatomical_side`, `anatomical_surface`, `anatomical_boundary` or other variants.

Composition of spatial terms

Use AT MOST ONE spatial term for each entity:

✓ Correct:

```
# Antero-lateral region of head red
this > aism-insect_head > bspo-antero_lateral_region >> pato-red;
```

✗ Incorrect:

```
# Antero-lateral region of head red
this > aism-insect_head > bspo-anterior_region > bspo-lateral_region >> pato-red;
```

Post-composition

Post-compose elementary anatomical structures to define more specific ones.

✓ Correct:

```
# Protibia with spur
this > aism-protibia > aism-cuticular_spur;
# Abdomen with 7 sternites
this > aism-abdominal_sternite .has_element_count 7;
```

✗ Avoid:

```
# Protibia with spur
this > aism-protibial_spur;
# Abdomen with 7 sternites
this > aism-abdomen_with_7_sternites;
```

Main types of phenotypic statement

1. `has_part` & `part_of`

The relationship `has_part` (alias: `>`) is used to state that an **entity** is part of another **entity**:

```
# Organism has head
this > aism-insect_head;
```

The reverse relationship `part_of` (alias: `<`) should be avoided unless strictly necessary to avoid less intelligible rendering:

```
# Head is part of organism
aism-insect_head < this;
```

2. `has_characteristic` & `inheres_in`

The relationship `has_characteristic` (alias: `>>`) is used to state that an **entity** is characterised by some **quality**:

```
# Head is red
this > aism-insect_head >> pato-red;
```

The reverse relationship `inheres_in` (alias: `<<`) should be avoided unless strictly necessary:

```
# Red inheres in head
pato-red << aism-insect_head < this;
```

3. `encircles` & `encircled_by`

The relationship `encircles` (alias: `->`) is used for **ring sclerites** connected via a conjunctiva (tarsus, tibia, femur, antennomeres, **setae**, etc.):

```
# Protibia with seta
this > aism-protibia -> aism-cuticular_seta;
```

The reverse relationship `encircled_by` (alias: `<-`) should be avoided unless strictly necessary:

```
# Seta is encircled by protibia
aism-cuticular_seta <- aism-protibia < this;
```

4. Absence

The absence of a structure can be stated by negating the `has_part` or `encircles` relationship via the negation operator (**!**):

```
# Antenna absent
this !> aism-antenna;
# Protibia without cuticular seta
aism-protibia !-> aism-cuticular_seta;
```

⚠️ **Do not** use the `pato-absent` quality:

```
# Antenna absent
this > aism-antenna >> pato-absent; # INCORRECT
```

5. Relative comparison

This pattern is used to compare the extents of two qualities, *i.e.*, if one is **larger than** or **smaller than** another. Used relationships:

- `increased_in_magnitude_relative_to` (alias: |>|)
- `decreased_in_magnitude_relative_to` (alias: |<|)

Patterns:

`'E1' >> 'Q1' |>| 'E2' >> 'Q2'` (larger than)
and
`'E1' >> 'Q1' |>| 'E2' >> 'Q2'` (smaller than)

Example:

```
# Protibia longer than protarsus
this > aism-protibia >> pato-length |>| pato-length << aism-protarsus < this;
```

The pattern can be generated automatically from the snippets by typing the **tmp** command and selecting it from the dropdown menu.

6. Relative measurements

Used to measure one quality using another as unit. It can be generated automatically with the **tmp** command.

```
relative_measurement:
  .measured_trait:
    .entity: aism-protibia:id-010498
    .quality: pato-length
  .unit:
    .entity: aism-protarsus:id-538e56
    .quality: pato-length
  .value: 3
```

Example:

```
# Protibia 3 times as long as protarsus
#>>>YAML relative_measurement
relative_measurement:
  .measured_trait:
    .entity: aism-protibia:id-010498
    .quality: pato-length
  .unit:
    .entity: aism-protarsus:id-538e56
    .quality: pato-length
  .value: 3
#<<<YAML
```

7. Absolute measurements

Allow you to provide a measurement value of a quality using standard measurement units (mm, µm, mg, etc.). Also generated through the **tmp** command.

```
absolute_measurement:
    .measured_entity: this
    .measured_quality: pato-length
    .value: 21.5
    .unit: unit-millimeter
```

Example:

```
# Body length: 21.5 mm
#>>>YAML absolute_measurement
absolute_measurement:
    .measured_entity: this
    .measured_quality: pato-length
    .value: 21.5
    .unit: unit-millimeter
#<<<\YAML
```

8. Element Count

States exact number of serial structures using `has_element_count` relationship:

```
# Protarsus with 5 protarsomeres
this > aism-protarsus > aism-protarsomere .phs-has_element_count 5;
```

9. Comparison of traits between different OTUs

This comparison is done with the `exclude` command and requires to have the descriptions of both organisms within the same file.

```
OTU = { # OTU: Scarabaeus viettei

  DATA = {
    #>>>YAML described_species
      described_species:
         .id: scarabaeus_viettei #OTU ID
         .rdfs-label: 'Scarabaeus viettei'
         .gbif_id: 'https://www.gbif.org/species/4997091'
         .dwc-Catalog_Number:
              - 'http://id.luomus.fi/GZ.15821'
          .is_a:
              - uberon-male_organism
              - uberon-adult_organism
              - dwc-Preserved_Specimen
      #<<<YAML
  }

  TRAITS = {

    # Protibia shorter than in Scarabaeus sakalava
    this > aism-protibia >> pato-length:id-111111 |<| pato-length:id-222222[exclude = TRUE];
  }
}

OTU = { # OTU: Scarabaeus sakalava

  DATA = {
    #>>>YAML described_species
    described_species:
        .id: scarabaeus_sakalava #OTU ID
        .rdfs-label: 'Scarabaeus sakalava'
        .gbif_id: 'http://zoobank.org/7AD8F87F-E7C1-4094-BD63-7662F167E9CB'
        .dwc-Catalog_Number:
            - 'http://id.luomus.fi/GZ.15827'
        .is_a:
            - uberon-male_organism
            - uberon-adult_organism
            - dwc-Preserved_Specimen
    #<<<YAML
  }

  TRAITS = {

    # Protibia longer than in Scarabaeus viettei
    this > aism-protibia >> pato-length:id-222222 |>| pato-length:id-111111[exclude = TRUE];
  }
}
```

10. Other useful relationships

- `coincident_with`

```
# Head grooved along posterior margin
this > aism-insect_head:id-56cc72 > aism-cuticular_groove .ro-coincident_with bspo-posterior_margin < aism-insect_head:id-56cc72;
```

- `medial_to`

```
# Head with tubercle placed medially to eye
this > aism-insect_head:id-2094bf > aism-cuticular_tubercle .aism-medial_to aism-eye_cuticle < aism-insect_head:id-2094bf;
```

- **Fused structures**

  To say that the protibial spur is fused with the protibia, you will need to use both the **relational quality** `fused_with` and the **relationship** `towards`:

  ```
  # Protibial spur fused with protibia
  this > aism-protibia:id-296644 > aism-cuticular_spur pato-fused_with:id-748ca8 .ro-towards aism-protibia:id-296644;
  ```

You may then want to specify the degree of fusion of the spur to the tibia (for example, completely/strongly fused). In this case, the quality `fused_to` is further specified as `increased_magnitude` through the relationship `has_modifier`:

```
# Protibia and spur completely fused
this > aism-protibia:id-296644 > aism-cuticular_spur pato-fused_with:id-748ca8 .ro-towards aism-protibia:id-296644;
pato-fused_with:id-748ca8 .ro-has_modifier pato-increased_magnitude;
```

Anatomical collections

In taxonomic descriptions, we often state the presence of an **unspecified** number of structures of the same kind (punctures, setae, hairs, scales, etc.). The simplest way could be using qualities such as `punctate` or `setose` but terms do not allow to characterize individual structures in further detail.

To do that, use `anatomical_collection` and, for a linear assemblage of structures, `anatomical_row`. The property `has_member` must be used when assigning entities to an anatomical collection.

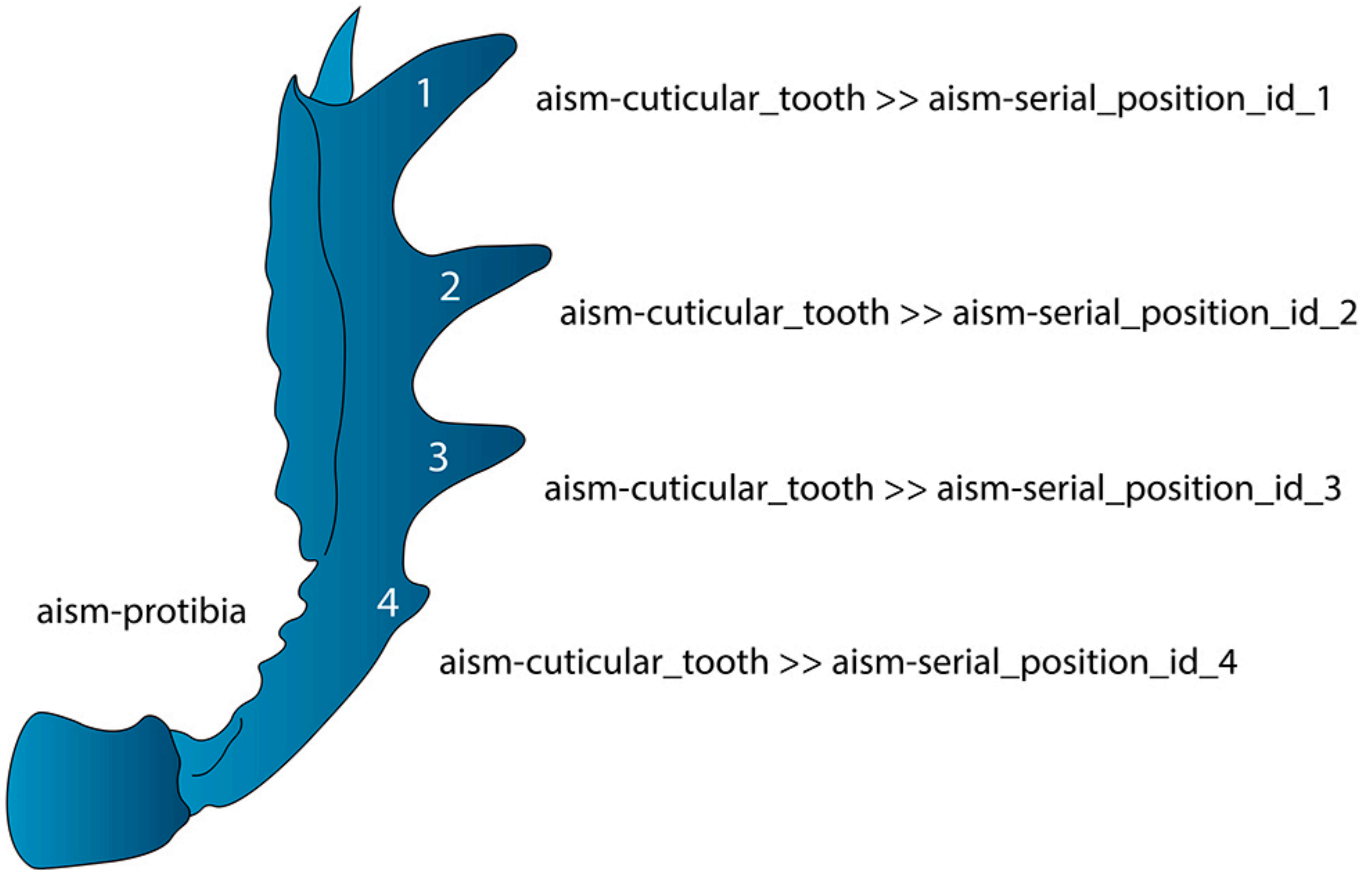
```
# Head with simple setigerous punctures
this > aism-insect_head > uberon-anatomical_collection .has_member aism-simple_setigerous_cuticular_puncture;
```

```
#Posterior margin of head with a row of setae
this > aism-insect_head > bspo-posterior_margin > uberon-anatomical_row .has_member aism-cuticular_seta;
```

Numbering serial structures

If you want to describe single structures within an ordinate series, you need to refer to their relative position within the series using the `aism-serial_position_id_X` quality.

```
# Dorsal margin of protibia with 4 cuticular teeth
this > aism-protibia > bspo-dorsal_margin > aism-cuticular_tooth >> aism-serial_position_id_1;
this > aism-protibia > bspo-dorsal_margin > aism-cuticular_tooth >> aism-serial_position_id_2;
this > aism-protibia > bspo-dorsal_margin > aism-cuticular_tooth >> aism-serial_position_id_3;
this > aism-protibia > bspo-dorsal_margin > aism-cuticular_tooth >> aism-serial_position_id_4;
```


1
2
3
4
aism-cuticular_tooth >> aism-serial_position_id_1
aism-cuticular_tooth >> aism-serial_position_id_2
aism-cuticular_tooth >> aism-serial_position_id_3
aism-protibia
aism-cuticular_tooth >> aism-serial_position_id_4

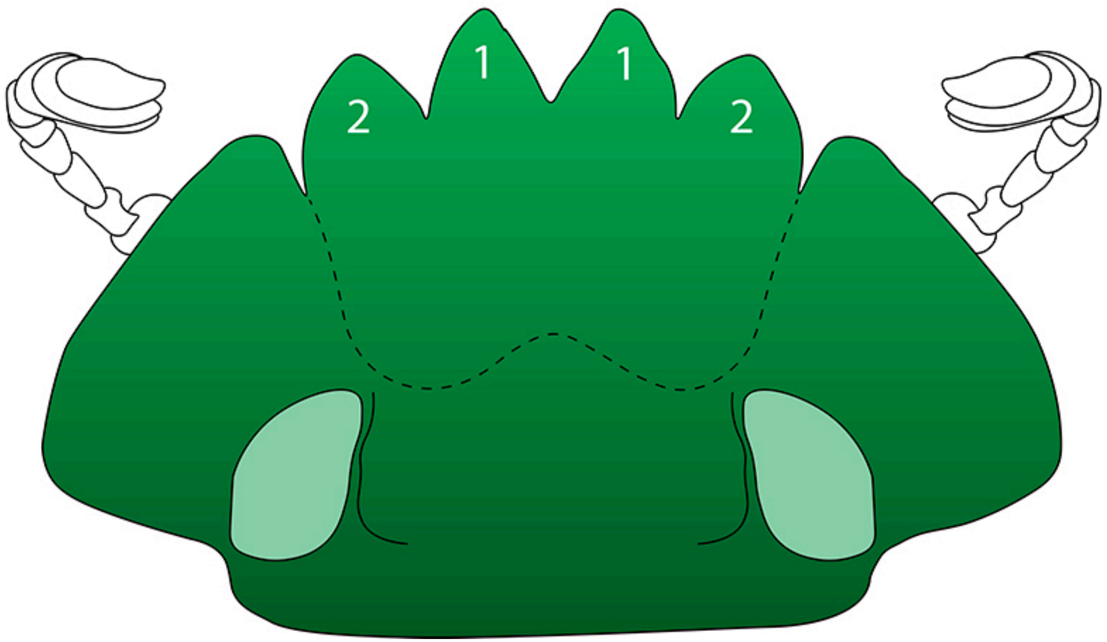
Numbering serial structures: bilateral pairs

For **bilaterally paired** structures, you don't need to specify the existence of both a left and right element. Use `bilaterally_paired` instead:

```
# Anterior clypeal margin with four teeth
this > aism-clypeus > bspo-anterior_margin > aism-cuticular_tooth >> (aism-serial_position_id_1, pato-bilaterally_paired);
this > aism-clypeus > bspo-anterior_margin > aism-cuticular_tooth >> (aism-serial_position_id_2, pato-bilaterally_paired);
```

1: aism-cuticular_tooth >> (aism-serial_position_id_1, pato-bilaterally_paired)

2: aism-cuticular_tooth >> (aism-serial_position_id_2, pato-bilaterally_paired)

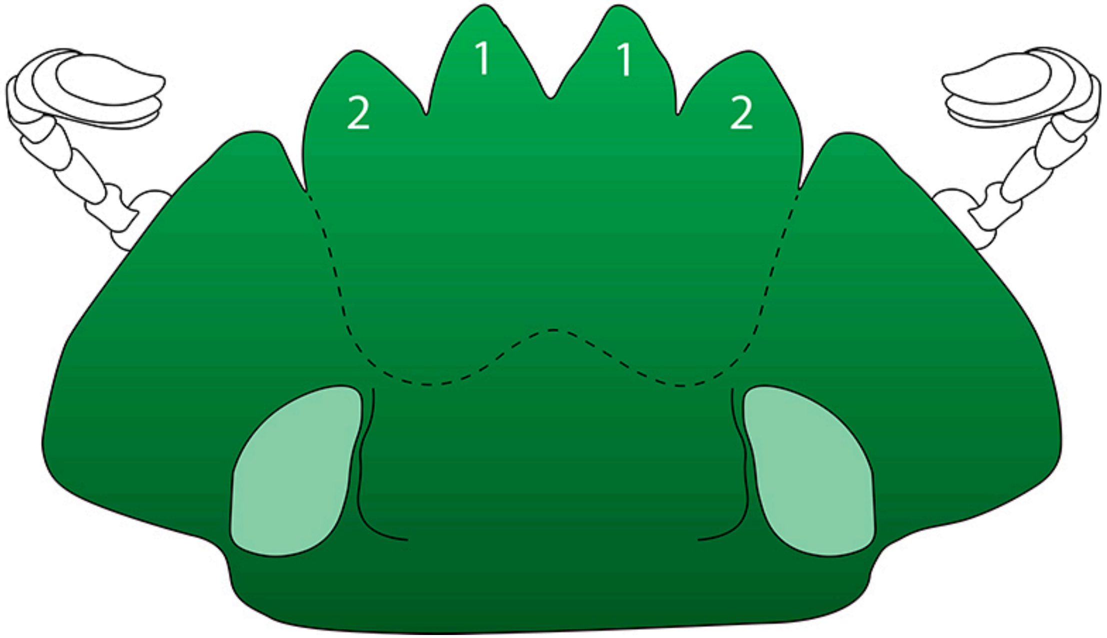


The position of each element relative to others can be further specified later, *e.g.*, teeth 1 are medial to teeth 2:

```
# Anterior clypeal margin with 4 teeth, teeth 1 medial to teeth 2
this > aism-clypeus > bspo-anterior_margin > aism-cuticular_tooth:id-tooth1 >> (aism-serial_position_id_1, pato-bilaterally_paired);
this > aism-clypeus > bspo-anterior_margin > aism-cuticular_tooth:id-tooth2 >> (aism-serial_position_id_2, pato-bilaterally_paired);
aism-cuticular_tooth:id-tooth1 .aism-medial_to aism-cuticular_tooth:id-tooth2;
```

1: aism-cuticular_tooth >> (aism-serial_position_id_1, pato-bilaterally_paired)

2: aism-cuticular_tooth >> (aism-serial_position_id_2, pato-bilaterally_paired)

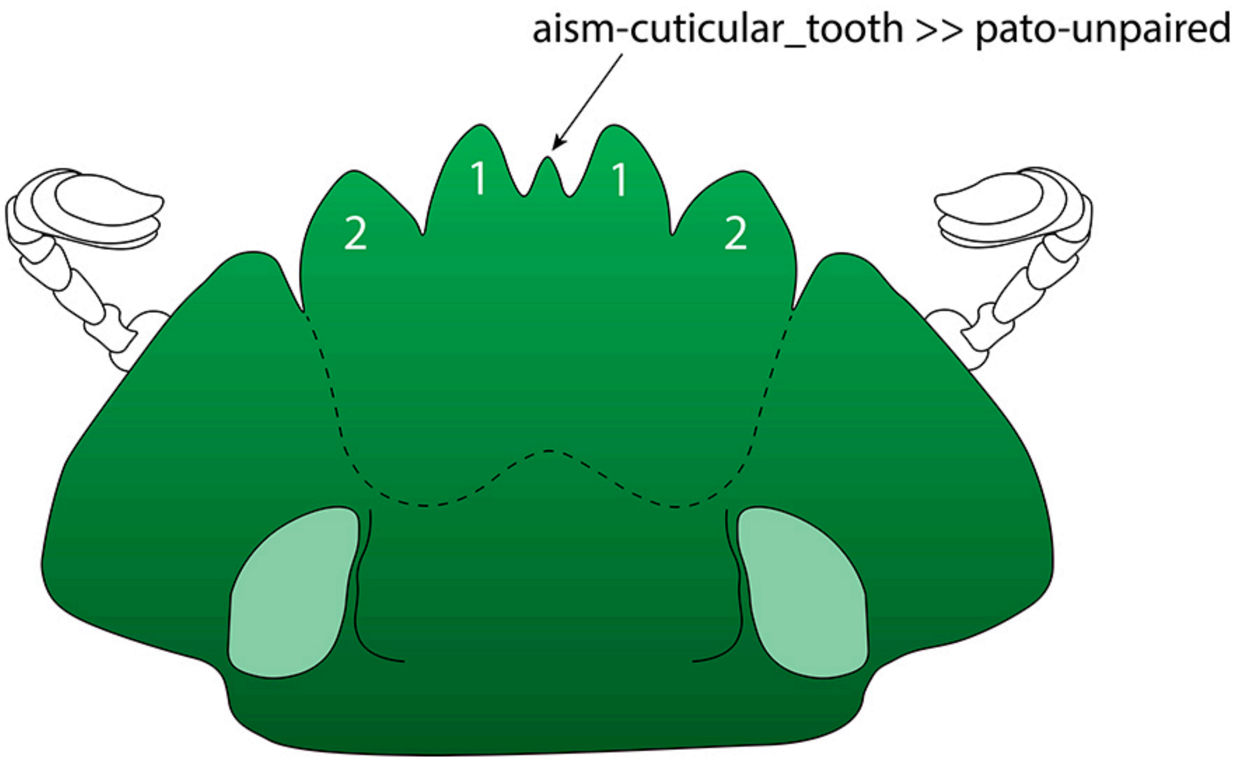


In the case of unpaired structures, the quality `unpaired` must be specified:

```
# Anterior clypeal margin with a single tooth
this > aism-clypeus > bspo-anterior_margin > aism-cuticular_tooth >> pato-unpaired;
```

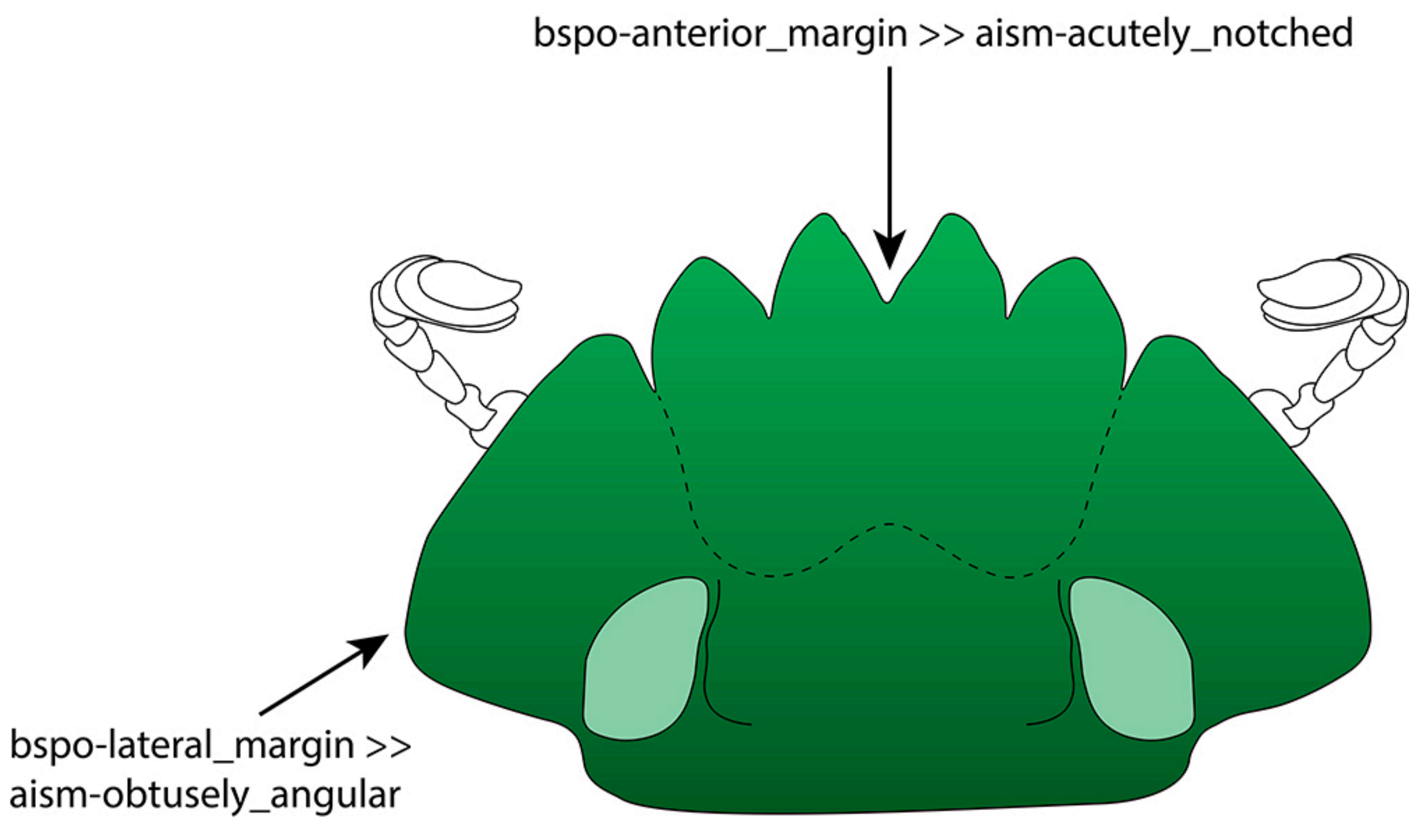
1: aism-cuticular_tooth >> (aism-serial_position_id_1, pato-bilaterally_paired)

2: aism-cuticular_tooth >> (aism-serial_position_id_2, pato-bilaterally_paired)



Some useful shape qualities

- **`angular`**: convexly angular margins;
  derivatives: `acutely_angular`, `obtusely_angular`, `right-angled`
- **`notched`**: concavely angular margins;
  derivatives: `acutely_notched`, `obtusely_notched`, `emarginate`


bspo-anterior_margin >> aism-acutely_notched
bspo-lateral_margin >>
aism-obtusely_angular

Limitations

Phenoscript suits most phenotypic descriptions. However, very complex structures (intricate color patterns, genital traits) are difficult to capture and may lead to intricate, poorly understandable statements.

In those cases, providing photographs instead of a textual description may be the best solution.

Phenoscript in under active development - suggestions of additional patterns are welcome!

Let's write a description!

Example description — *Scarabaeus* (Madagascar):

whs-phenoscript-2026/examples/Scarabaeus/phenotypes/scarabaeus_madagascar.yphs

Based on doi.org/10.3897/BDJ.12.e121562